Another (related) category is distributed query engines, which can, among other things, use columnar data formats. These include

- PrestoDB (https://prestodb.io/), open sourced by Facebook (Meta)
- Cloud provider offerings, such as Amazon Redshift (https://aws.amazon.com/redshift/) and Google BigQuery (https://cloud.google.com/bigquery)

In the last part of this chapter, we shift gears and move to a hot topic: out of those many choices, how to best go about selecting one (or more) backends.

5.6 Selecting backend destinations

Selecting backends can be tough. To give you a feeling for the sheer number of options, have a look at the CNCF landscape in the Observability and Analysis (https://landscape.cncf.io/card-mode?category=observability-and-analysis) space. It sports many different options, from open source to SaaS.

Now that you have an idea about the spectrum of offerings in the backends domain, let's turn our attention to picking a backend that is a good fit for your needs. Working backward from your workload and your requirements is something I recommend. Also, be aware that it is not uncommon to end up using multiple types of backends in concert. For example, you might use CNCF Cortex for metrics LTS and federation along with OpenSearch for log and trace storage and query. Finally, remember that certain frontends, such as Grafana, as the canonical example, allow you to bring together signals from various backends, effectively building out a single pane of glass for your observability needs. We will discuss this in detail in chapter 6.

We discussed return on investment (ROI) in earlier chapters, and with backends, this is of central importance. Let's dive into the costs aspect first.

5.6.1 Costs

In the ROI context, backends require investment, typically in the mid- to long-term range—that is, years. This is due to the fact that data has gravity, meaning wherever the majority of your data resides defines your primary environment (think on-prem versus cloud or between different cloud providers). It costs time and money to transfer data between environments, and you want to avoid unnecessary egress and ingress data transfer costs.

More specifically, the monetary investments for backends can break down into

- *Ingestion costs*—An agent, like the OpenTelemetry collector, ingests (or writes) signals into the backend. For example, for a Prometheus-compatible backend, this may be measured in USD per million samples ingested.
- *Storage costs*—After ingestion, signals need to be stored for long-term retrieval. These costs are usually not the biggest contributor to your bill (most often ingestion or query is).

  Concerning storage, one often useful property is the "retention period." With this, your provider expresses how long your signal will be stored and is available